

Question Paper

Exam Date & Time: 29-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [IT_2123]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Simplify a 4-variable Boolean function using K-map and implement using NAND gates only. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Application"
B) Design a circuit for $F(A,B,C,D)=\Sigma(0,1,2,5,8,9,10)$ using 4-variable K-map. (3)
Course Outcome: "CO1" / Cognitive Level: "Application"
C) Differentiate between signed and unsigned addition with a numerical example. (2)
Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Explain the design of a BCD adder and how it differs from a binary adder. (4)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
B) Implement a 16-to-1 Multiplexer using logic gates and derive the Boolean expression. (3)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
C) Construct an arithmetic comparison circuit to compare two 2-bit numbers. (3)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Develop a Verilog code for a 4-bit Ring Counter (3)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Differentiate between ripple counter and synchronous counter in terms of timing. (4)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Design a 4-bit synchronous counter using D flip-flops. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Design a sequence controller for Booth's multiplier. (5)
A) Course Outcome: "CO4" / Cognitive Level: "Analysis"

B) Evaluate the impact of different cache replacement strategies on system efficiency. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

C) Apply the concept of addressing modes to write an instruction using immediate addressing. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

5) Compare the characteristics of CISC and RISC architectures with examples. (5)

A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

B) Explain how pipelining improves performance and discuss hazards in pipelines. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

C) Compare hardwired and microprogrammed control units based on flexibility and speed. (2)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_2123:DIGITAL SYSTEMS AND COMPUTER ORGANIZATION

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_2123:DIGITAL SYSTEMS AND COMPUTER ORGANIZATION

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	8	26(52%)
4	Analysis	7	24(48%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)